

Lab05 - PCB

MTRN3100 - UNSW School of Mechanical and Manufacturing Engineering

Prerequisites

This lab is based on EAGLE 9.6.2 education which is free to download: [here](#)

Eagle 6.6 is available through [myaccess](#). While the interface is graphically different this lab can still be completed.

Introduction - Workflow

To make a PCB we just have to complete these 4 tasks.

- Create a library
- Create a schematic
- Design the layout
- Export and manufacture

Pre-made libraries

Download and import the spark fun eagle library found [here](#). Extract the file and place it in the libraries section in Eagle. Make sure to activate the libraries by clicking the grey dots to turn them green. Otherwise, they will not show up.

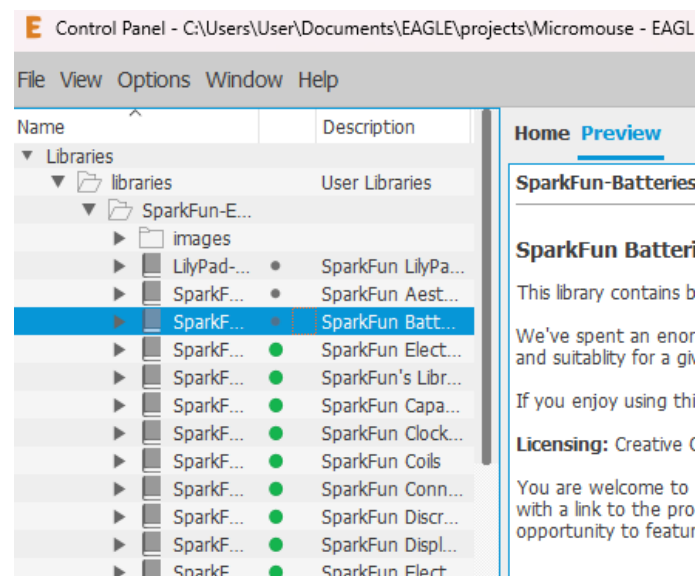


Figure 1: Sparkfun eagle libraries

Task 1 - Create a library

First, let's create a library to store the components we will use for the MTRN3100 Micromouse. From the control panel select:

File – > New – > Library

Then:

File – > Save as

And call your library MTRN3100. Make sure to activate it by selecting the grey dot in the control panel. This should turn it green.

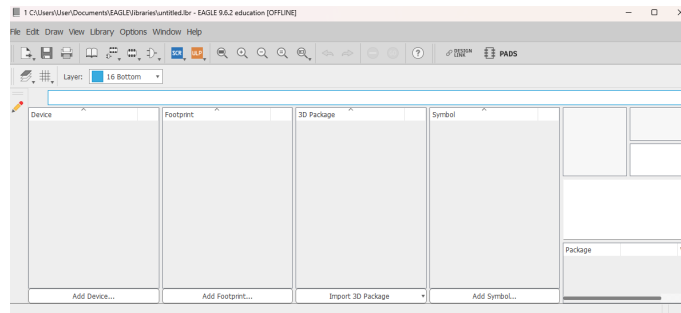


Figure 2: Library table of contents

Task 2 - Import existing components

Some components already exist in libraries. This is always a good thing to check before resorting to creating the component.

LED

Let's get an LED by searching for the component. Note when searching for components which you don't know the exact name of use * either side of the word to get all items containing the text, EG: *LED*.

Add Device – > Import – > Search for “LED5MM” – > Select LED5MM – > OK

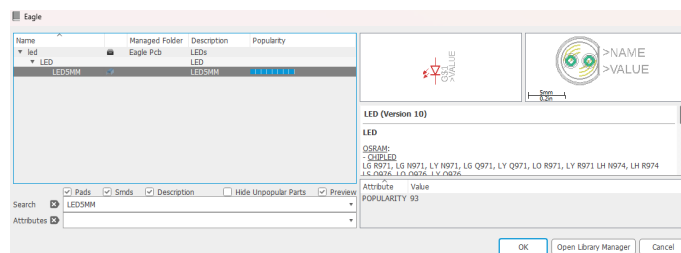


Figure 3: LED5MM component selected

To navigate back to the table of contents select the table of contents button .

Header pins

Alternatively, you can browse a library to see if it already has the correct component.

Add Device – > Import – > Scroll down and select the library “con-1stb” – > Find MA04-1 – > OK

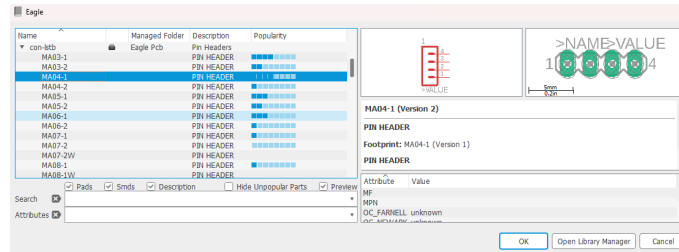


Figure 4: Header Pin component selected

Task - Additional Items

Have a go at adding additional items:

- DC Jack - DCJ0202 (Please note: This is not the DC jack from the kits)
- Resistor - AXIAL-0.3 (Spark fun library)
- Motor Terminal Block - MPT6

Task 3 - Create a new component

Certain components may not be in a library and we will have to make them ourselves. Let's use the motor driver as an example. Open the attached motor driver datasheet as it will be required as a reference. First, we need to create the physical footprint for our component. Then the symbol that is used in the schematic. Finally, they can be linked together, creating our device.

NOTE: The reason PCB software is set up like this is that you may have the same component and as such diagram, but with a different footprint.

Create Footprint

1. Create the footprint: **Add Footprint...** - > **Name it "MOTORDRIVER"** - > **OK**
2. Click on grid settings icon  and change the grid size to 100 mil.
3. Click on the through hole icon  and place a pad on the grid close to the mark  (It does not have to be directly on the mark)
4. Click on the information icon  then select a pad to open the settings menu. Change the shape to round. Additionally, We need to change the hole size so that the pin can fit through. Change the Drill to 45 (Note: this is in mil not mm, If you want it in mm you need to set it in the grid setting). If you would like to increase the size of the metal pad change the diameter setting.
5. Create pads to match the datasheet. You can copy and paste items additionally, move them with the move button 
6. Click on grid settings icon  and change the grid size to 50 mil.
7. Click on the draw line icon  and draw a box matching the datasheet around the pads.
8. Click on text icon . Place text saying ">VALUE" and ">NAME" anywhere around your component, These are variables that will be replaced automatically later. Add text saying GND near the GND pins. NOTE: Right-click on the mouse will rotate the text 90 degrees.
9. Click the name icon  then select each pin and change the name of it to match the datasheet. Eagle does not like pins having the same name, to fix this name the second GND pin GND@1. Eagle will remove all the symbols after the @ symbol and treat the pin as a GND pin.
10. Great work! Save your file and return to the table of contents.

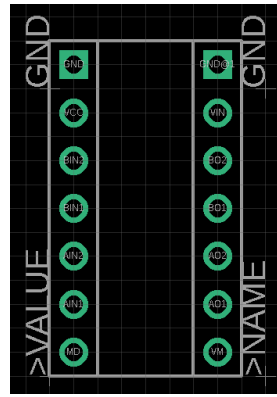


Figure 5: Example footprint of the motor driver.

Create Symbol

1. Create the symbol: **Add Symbol...** - > **Name it "MOTORDRIVER"** - > **OK**
2. Click on grid settings icon  and change the grid size to 100 mil.
3. Click on the pin icon  and place a pin on the workspace.
4. Click on the information icon  then select a pin to open the settings menu. Change the pin length to short.
5. Place 14 pins in a similar layout to the footprint. Select all 7 pins on one side and mirror them using the mirror tool .
7. Click on draw line icon  and draw a box connecting the pins.
8. Click on text icon . A text saying ">VALUE" and ">NAME" anywhere around your component, These are variables that will be replaced automatically later. NOTE: Right-click on the mouse will rotate the text 90 degrees.
9. Click the name icon  then select each pin and change the name of it to match the datasheet. Eagle does not like pins having the same name, to fix this name the second GND pin GND@1. Eagle will remove all the symbols after the @ symbol and treat the pin as a GND pin.
10. Great work! Save your file and return to the table of contents.

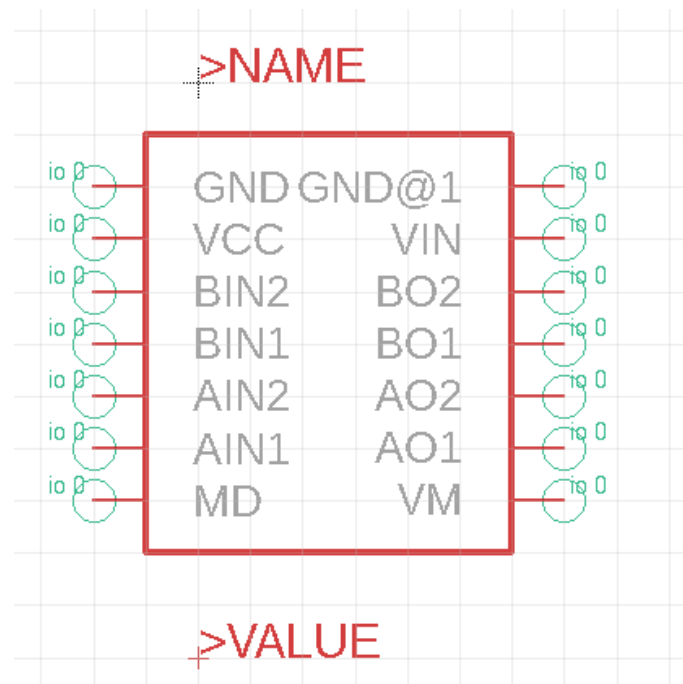


Figure 6: Example symbol of the motor driver.

Create Device

1. Create the device: **Add Device...** –> **Name it “MOTORDRIVER”** –> **OK**
2. Click on add icon  and add the symbol to the device.
3. Add the footprint: **New** –> **Add local package** –> **MOTORDRIVER**
4. Select: **Connect**
5. If you have correctly set up your pin names then select the corresponding pin and pad names and hit **Connect** to bind them.
6. Great work! Save your file and return to the table of contents.

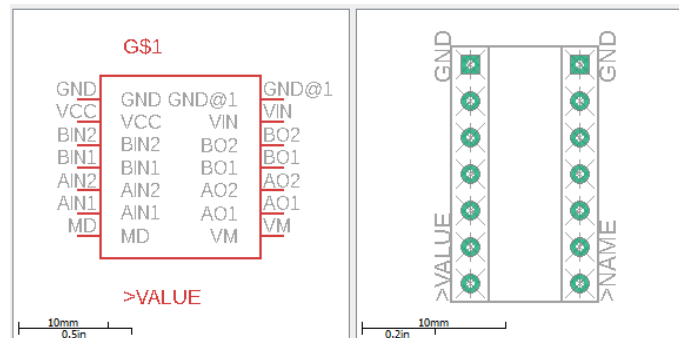


Figure 7: Finished motor driver component.

Practice Tasks

Datasheets for the Switch, Fuse, and Arduino have been provided. Try adding these components to your MTRN3100 library.

Task 4 - Schematic

Now, let's create a project and then a schematic to connect the components from our library together. From the control panel:

File –> **New** –> **Project** –> **name it “Micromouse”**

Activate the project by selecting the grey dot. Then:

Right click Micromouse –> **New** –> **Schematic** –> **File** –> **Save as** –> **name it “Micromouse”**

1. Click on add icon  and scroll down to find the MTRN3100 library.
2. Add: 1x MOTORDRIVER, 2x MPT6, 1x AXIAL-0.3, 1x LED5MM, 1x DCJ0202, 1x MA04-1.
3. Rearrange the components by moving, rotating, and mirroring them to match the example shown in Figure 8.
4. Using the net tool  connect the pins to match the example shown in Figure 8.
5. Using the value icon  select the resistor and set the value to 1.2K.
6. Great work! Save your file.

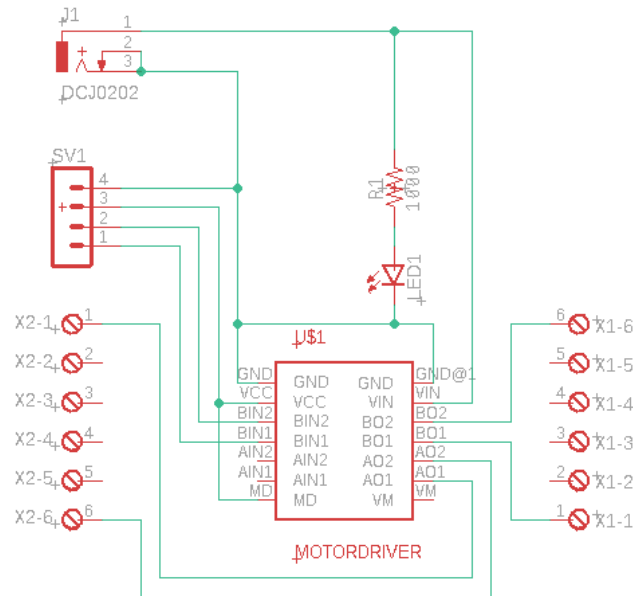


Figure 8: Example schematic

Task 5 - Board

1. Click on schematic to board icon  .
2. Move components on the board by clicking the + icon on each component and dragging them. Press right-click to rotate the item. Try to place the components logically and with as few cross-overs.
3. Select the walls of the board and bring in the board dimensions.

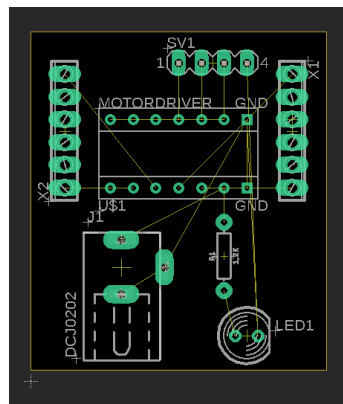


Figure 9: Example of an un-optimised board (before step 4).

4. Select the ratsnest icon  to clean the wires.
5. Routing time! Autorouting can be used by selecting  however may not always produce the desired result.
6. To undo a wire select rip up and click the wire .
7. To manually route a wire select route  use left and right click to create the desired shapes. Via points can be added using .



Figure 10: Route settings.

8. Add extra holes for mounting using the hole command . Just be aware the units for the hole size are specified in the grid setting. Add extra using text the text command  or other silk screen drawing commands, also remove any unwanted text.

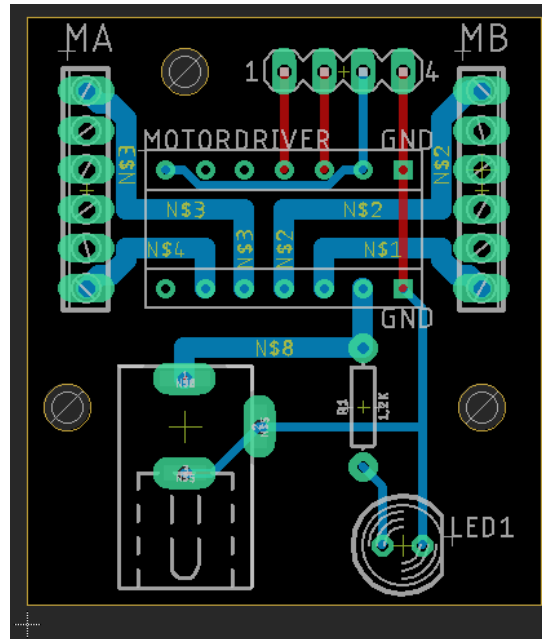


Figure 11: Partial development of Micromouse PCB. (BLUE: Bottom layer, RED: Top Layer)

Task 6 - Conformation and Export

1. It is good practice to check your design for errors. One way to do this is to select and run the design check . You may need to scroll down for this.
2. Click on the manufacturing tab to see a render of your PCB.
3. Export your CAM files by pressing . All that is left to do is submit your files to a PCB manufacturer and then wait. Hopefully, you didn't make any mistakes.

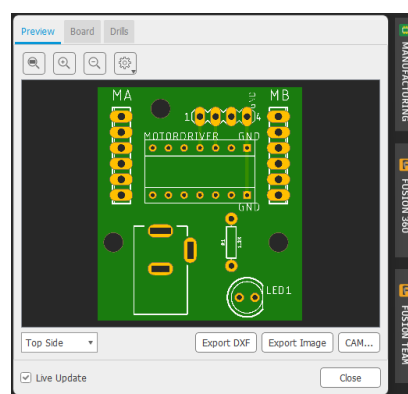


Figure 12: Example of manufacturing render.

What's Next

Hopefully, this lab has provided you with all the skills you need to make your own Micromouse PCB. The complete set of component datasheets has been provided with the project. Use these to create new components. Additionally, the wiring schematic can also be used as a reference but you are free to make your own.

If you want to learn more about PCB design this tutorial only scratched the surface. Ground planes can be a useful tool for routing and more complex boards. Additionally, any monochrome image can be added to the silk screen. Happy designing!